AI-Driven Optimization: From Algorithms to Applications

Xianchao Xiu

Department of Automation



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Joint work with Jingjing Liu (SHU), Hui Chen (SUEP), Wanquan Liu (SYSU) and others

Outline

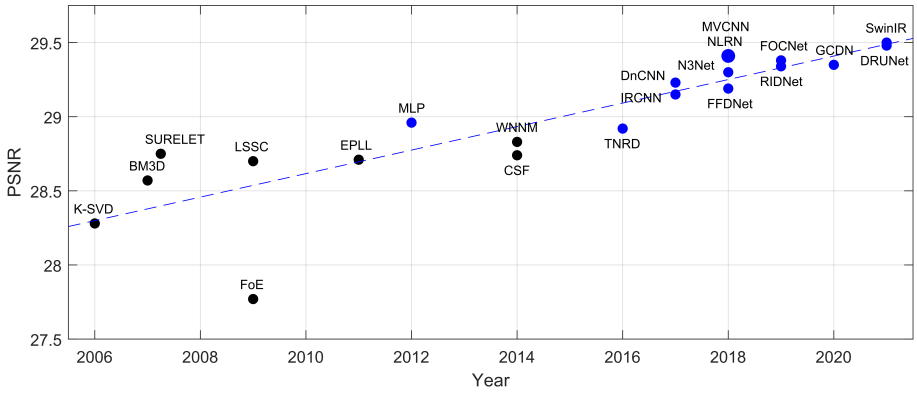
Introduction

Recent Advances

Future Work

Why

- Elad-Kawar-Vaksman, Image Denoising: The Deep Learning Revolution and Beyond, SIIMS, 2023



How

- Iterative shrinkage thresholding algorithm (ISTA)

$$\min_{x\in\mathbb{R}^n} \quad \frac{1}{2}\|Ax-b\|^2+\lambda\|x\|_1$$

  - Step 1: Compute gradient

$$r^{k+1}=x^k-\rho A^\top(Ax^k-b)$$

  - Step 2: Compute shrinkage

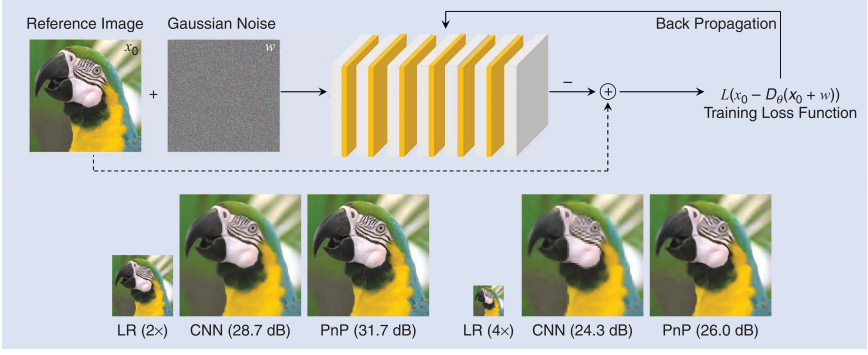
$$\begin{aligned} x^{k+1} &= \arg\min_x \ \frac{1}{2}\|x-r^{k+1}\|^2+\lambda\|x\|_1 \\ &= \mathtt{prox}_{\lambda\|\cdot\|_1}(x^k-\rho A^\top(Ax^k-b)) \end{aligned}$$

- Two important issues need to be considered
  - How to design regularizations $\Rightarrow$ Deep PnP/Deep Unrolling
  - How to choose parameters $\Rightarrow$ Deep Unrolling

Deep PnP

- Venkatakrishnan-Bouman-Wohlberg, Plug-and-Play Priors for Model Based Reconstruction, IEEE GlobalSIP, 2013

$$\min_x \quad \frac{1}{2}\|Ax-b\|^2+\lambda\|x\|_1 \quad \Rightarrow \quad \min_x \quad \frac{1}{2}\|Ax-b\|^2+\lambda\phi(x)$$



Related Works

- Ryu-Liu-Wang et al, Plug-and-Play Methods Provably Converge with Properly Trained Denoisers, ICML, 2019
- Wei-Rivero-Liang et al, Tuning-free Plug-and-Play Proximal Algorithm for Inverse Imaging Problems, ICML, 2020
- Sun-Wu-Xu et al, Scalable Plug-and-Play ADMM With Convergence Guarantees, IEEE TCI, 2021
- Cohen-Elad-Milanfar, Regularization by Denoising via Fixed-Point Projection (RED-PRO), SIIMS, 2021
- Huraul-Leclaire-Papadakis, Proximal Denoiser for Convergent Plug-and-Play Optimization with Nonconvex Regularization, ICML, 2022
- Tan-Mukherjee-Tang et al, Provably Convergent Plug-and-Play Quasi-Newton Methods, SIIMS, 2024
- Wang-Zhang-Li et al, DMPlug: A Plug-in Method for Solving Inverse Problems with Diffusion Models, NeurIPS, 2024
- Zheng-Chu-Zhang et al,InverseBench: Benchmarking Plug-and-Play Diffusion Priors for Inverse Problems in Physical Sciences, ICLR, 2025

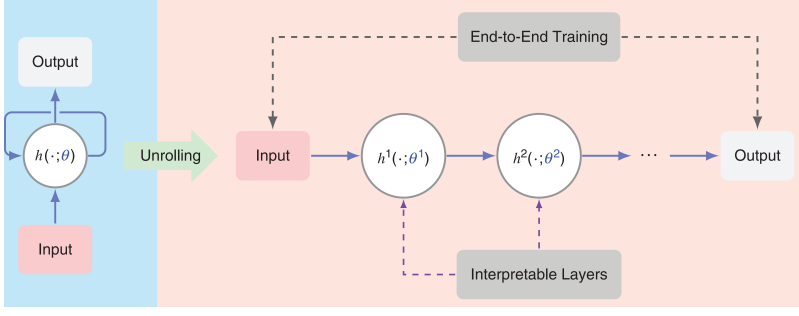
Deep Unrolling

- Gregor-LeCun, Learning Fast Approximations of Sparse Coding, ICML, 2010

$$x^{k+1} = \mathtt{prox}_{\lambda\|\cdot\|_1}(x^k - \rho A^\top (Ax^k - b))$$

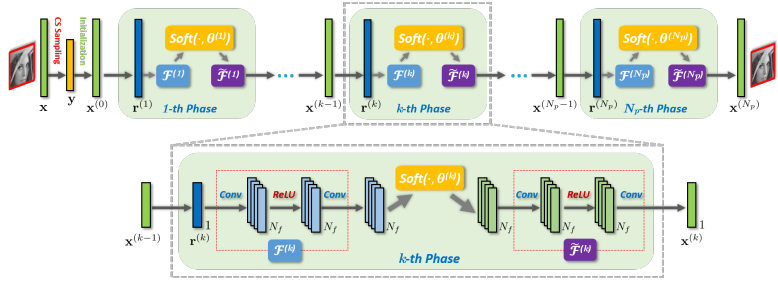
$$\Downarrow$$

$$x^{k+1} = h_{\theta^k}(x^k - \rho A^\top (Ax^k - b))$$



Deep Unrolling

- Zhang-Ghanem, ISTA-Net: Interpretable Optimization-Inspired Deep Network for Image Compressive Sensing, CVPR, 2018

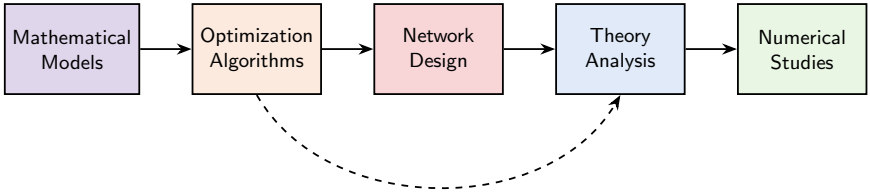


Algorithm	CS Ratio							Time
	50%	40%	30%	25%	10%	4%	1%	CPU/GPU
TVAL3 [22]	33.55	31.46	29.23	27.92	22.99	18.75	16.43	3.135s/———
D-AMP [28]	35.92	33.56	30.39	28.46	22.64	18.40	5.21	51.21s/———
IRCNN [49]	36.23	34.06	31.18	30.07	24.02	17.56	7.70	———/68.42s
SDA [30]	28.95	27.79	26.63	25.34	22.65	20.12	17.29	———/0.0032s
ReconNet [21]	31.50	30.58	28.74	25.60	24.28	20.63	17.27	———/0.016s
ISTA-Net	37.43	35.36	32.91	31.53	25.80	21.23	17.30	0.923s/0.039s
ISTA-Net$^+$	**38.07**	**36.06**	**33.82**	**32.57**	**26.64**	**21.31**	**17.34**	1.375s/0.047s

Algorithm	CS Ratio				Time
	20%	30%	40%	50%	GPU
ADMM-Net	37.17	39.84	41.56	43.00	0.9535s
ISTA-Net	38.30	40.52	42.12	43.60	0.1246s
ISTA-Net$^+$	**38.73**	**40.89**	**42.52**	**44.09**	0.1437s

Related Works

- Chen-Liu-Wang et al, Theoretical Linear Convergence of Unfolded ISTA and Its Practical Weights and Thresholds, NeurIPS, 2018
- Yang-Sun-Li et al, ADMM-CSNet: A Deep Learning Approach for Image Compressive Sensing, IEEE TPAMI, 2020
- Song-Zhang-Zhou et al, SOR-Unrolled Learning-to-Solve Method for Absolute Value Equations, JORSC, 2026
- Oshin-Ghosh-Saravanos et al, Deep FlexQP: Accelerated Nonlinear Programming via Deep Unfolding, ICLR, 2026
- Xiu-He-Qu et al, A Survey of Deep Unfolding Methods for Image Compressive Sensing (in Chinese), ORT, 2026



Outline

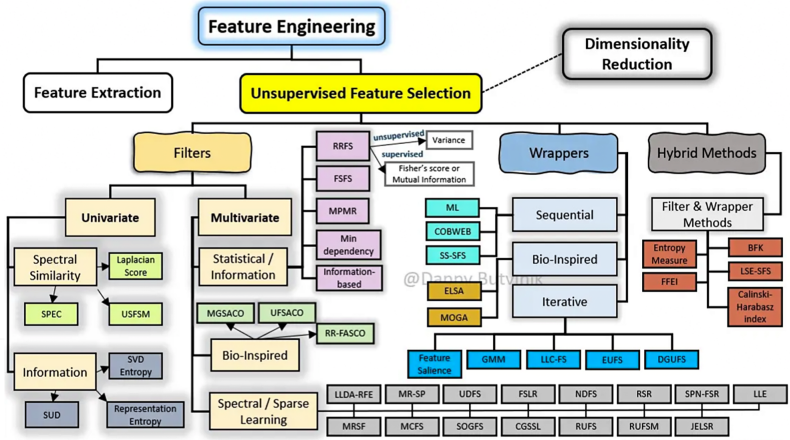
Introduction

Recent Advances

Future Work

Part I

- Unsupervised feature selection *vs.* Feature extraction
- Select a subset of input features without labels



PCA

- Given $X = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n) \in \mathbb{R}^{d \times n}$, principal component analysis (PCA) is

$$\begin{aligned} \min_{W \in \mathbb{R}^{d \times p}} \quad & -\operatorname{Tr}(W^\top X X^\top W) \\ \text{s.t.} \quad & W^\top W = I \end{aligned}$$

- Li-Nie-Bian et al, Sparse PCA via $\ell_{2,p}$-Norm Regularization for Unsupervised Feature Selection, IEEE TPAMI, 2023

$$\begin{aligned} \min_{W} \quad & -\operatorname{Tr}(W^\top X X^\top W) + \lambda \|W\|_{2,p}^p \ (0 < p < 1) \\ \text{s.t.} \quad & W^\top W = I \end{aligned}$$

- Xiu-Huang-Shang et al, Bi-Sparse Unsupervised Feature Selection, IEEE TIP, 2025

$$\begin{aligned} \min_{W} \quad & -\operatorname{Tr}(W^\top S W) + \lambda_1 \|W\|_{2,p}^p + \lambda_2 \|W\|_q^q \ (0 \le p, q < 1) \\ \text{s.t.} \quad & W^\top W = I \end{aligned}$$

Model

- Liu-Tai-Xiu et al, Learning to Select Features with Manifold Deep Unfolding PCA, submitted to IEEE TIP

$$\begin{aligned} \min_{X} \quad & \frac{1}{2}\|A - XX^\top A\|_{\mathrm{F}}^2 + \lambda\|X\|_{2,0} + \mu\|X\|_0 \\ \text{s.t.} \quad & X^\top X = I \end{aligned}$$

- Alternating direction method of multipliers (ADMM)

$$\begin{aligned} \min_{X,Y,Z} \quad & f(X) + \lambda\|Y\|_{2,0} + \mu\|Z\|_0 \\ \text{s.t.} \quad & X^\top X = I,\ X = Y,\ X = Z \end{aligned}$$

$$\Downarrow$$

$$\begin{aligned} \mathcal{L}(X, Y, Z, \Lambda, \Pi) = & f(X) + \lambda\|Y\|_{2,0} + \mu\|Z\|_0 \\ & + \langle \Lambda, X - Y\rangle + \frac{\alpha}{2}\|X - Y\|_{\mathrm{F}}^2 + \langle \Pi, X - Z\rangle + \frac{\beta}{2}\|X - Z\|_{\mathrm{F}}^2 \end{aligned}$$

Algorithm

- **Input:** Data A, stage-wise parameters $\lambda_k, \mu_k, \alpha_k, \beta_k, \eta_k$
- **Initialize:** $X^0, Y^0, Z^0, \Lambda^0, \Pi^0$
  - Update X^{k+1} according to

$$X^{k+1} = \texttt{MGradNet+}(X^k, Y^k, Z^k, \Lambda^k, \Pi^k, G, \alpha_k, \beta_k, \eta_k)$$

  - Update Y^{k+1} according to

$$Y^{k+1} = \texttt{HardNet}(U^k, \tau_k)$$

  - Update Z^{k+1} according to

$$Z^{k+1} = \texttt{EHardNet}(V^k, \gamma_k)$$

  - Update Λ^{k+1} according to

$$\Lambda^{k+1} = \texttt{Linear}(\Lambda^k, X^{k+1}, Y^{k+1}, \alpha_k)$$

  - Update Π^{k+1} according to

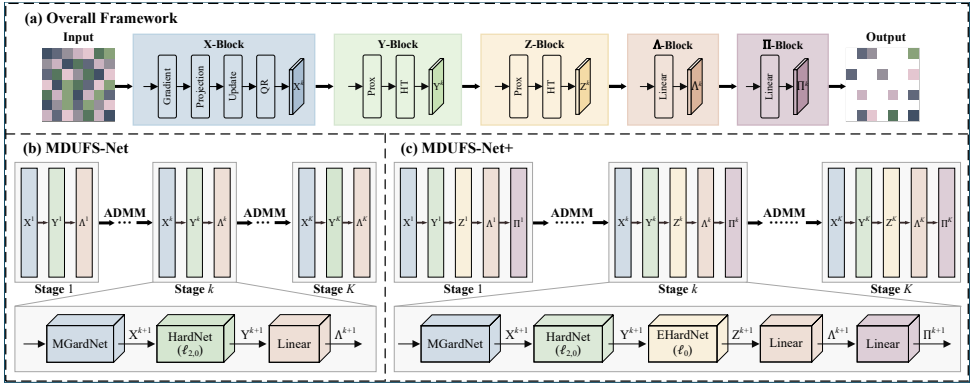
$$\Pi^{k+1} = \texttt{Linear}(\Pi^k, X^{k+1}, Z^{k+1}, \beta_k)$$

- **Output:** X^K, Y^K, Z^K

Architecture

- All papameters $(\lambda, \mu, \alpha, \beta)$ are trained in an end-to-end manner
- The loss is defined as

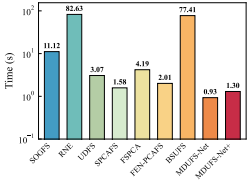
$$\text{Loss} = f(X) + \lambda\varphi_{2,0}(Y) + \mu\varphi_0(Z)$$



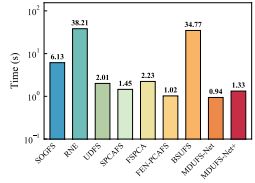
Results

Methods	COIL20	ISOLET	USPS	PIE	LUNG	MSTAR	MNIST	HAR	Average
ALLfea	64.73±7.63 (10)	59.87±2.78 (10)	59.82±1.29 (10)	27.43±0.96 (10)	72.57±7.63 (10)	90.29±3.94 (10)	47.69±2.78 (10)	53.79±0.99 (10)	59.52±3.50
LapScore	55.36±1.58 (90)	54.07±1.80 (100)	52.31±2.68 (100)	33.90±0.97 (30)	58.40±0.75 (100)	68.67±3.32 (100)	48.62±0.15 (80)	61.59±0.09 (100)	54.12±1.42
SOGFS	43.32±2.96 (100)	43.43±1.84 (30)	45.23±1.31 (100)	24.73±0.94 (100)	54.02±3.19 (100)	64.65±2.66 (60)	41.26±1.15 (80)	46.05±0.38 (100)	45.34±1.80
RNE	56.64±3.40 (80)	50.29±1.77 (100)	57.51±2.01 (100)	22.39±0.61 (60)	58.90±5.79 (100)	69.11±1.02 (80)	39.58±0.19 (100)	42.16±0.03 (60)	49.57±1.85
UDFS	48.38±2.41 (100)	40.84±1.83 (100)	50.63±1.93 (100)	27.49±0.94 (100)	58.19±5.29 (100)	75.24±3.37 (90)	34.57±1.37 (80)	42.03±0.03 (100)	47.17±2.15
SPCAFS	45.72±2.07 (80)	40.55±1.76 (100)	57.95±1.57 (80)	45.41±1.74 (80)	56.32±4.29 (90)	67.96±2.81 (100)	31.71±0.91 (90)	58.67±0.01 (90)	50.54±1.90
FSPCA	56.57±1.89 (80)	47.99±1.31 (50)	56.05±1.96 (60)	32.45±1.18 (80)	55.74±1.90 (90)	80.09±5.55 (70)	49.49±1.43 (70)	44.54±0.05 (40)	52.87±1.91
FEN-PCAFS	58.20±2.24 (100)	52.28±2.11 (100)	58.28±2.31 (100)	38.94±1.64 (80)	61.77±1.53 (80)	81.07±3.84 (80)	47.66±0.02 (60)	60.91±0.04 (20)	57.39±1.72
BSUFS	57.26±2.37 (100)	53.59±2.12 (100)	59.33±0.96 (80)	45.25±1.74 (80)	60.13±2.42 (70)	71.45±1.81 (100)	33.80±0.89 (100)	63.27±0.05 (90)	55.51±1.55
MDUFS-Net	**65.60±2.83 (100)**	62.84±2.27 (100)	62.02±1.69 (100)	45.61±1.73 (70)	66.12±1.86 (100)	88.00±1.83 (90)	56.10±0.05 (100)	67.53±0.25 (40)	64.23±1.56
MDUFS-Net+	65.23±1.89 (90)	**64.16±2.04 (100)**	**64.79±1.65 (80)**	**46.82±1.65 (60)**	**69.46±1.35 (60)**	**91.70±0.15 (100)**	**56.75±0.03 (100)**	**70.19±0.13 (20)**	**66.14±1.11**

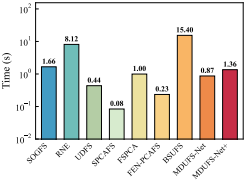
Runtime



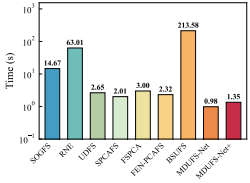
(a) COIL20



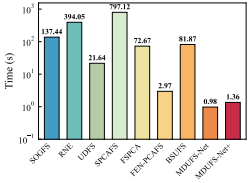
(b) ISOLET



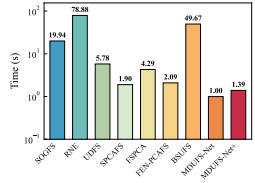
(c) USPS



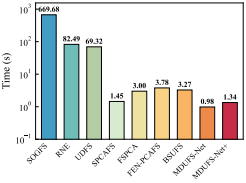
(d) PIE



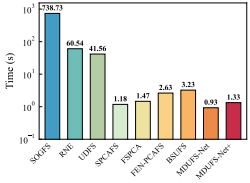
(e) LUNG



(f) MSTAR



(g) MNIST

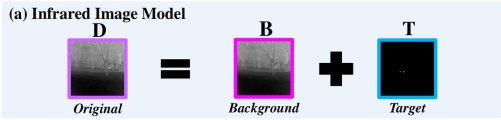


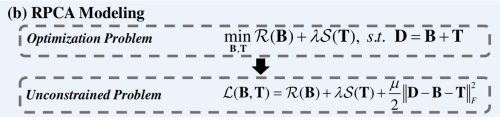
(h) HAR

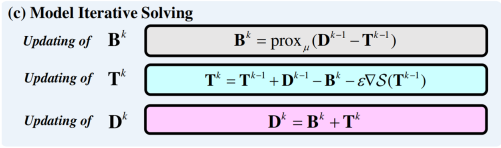
Part II

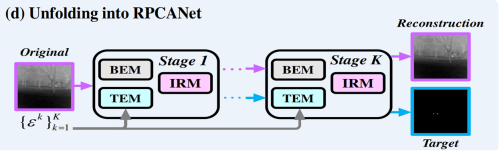
- Wu-Zhang-Li et al, RPCANet: Deep Unfolding RPCA Based Infrared Small Target Detection, WACV, 2024

$$\min_{B,T} \quad \|B\|_* + \lambda \|T\|_1$$
$$\text{s.t.} \quad D = B + T$$









Model

- Liu-Han-Xiu et al, Lightweight Deep Unfolding Networks with Enhanced Robustness for Infrared Small Target Detection, IEEE TIP, AQ

$$\begin{aligned} \min_{B,T} \quad & \|B\|_* + \lambda \|T\|_1 \\ \text{s.t.} \quad & D = B + T \end{aligned}$$

$$\Downarrow$$

$$\begin{aligned} \min_{B,T,N} \quad & \|B\|_* + \lambda \|T\|_1 + \mu \|N\|_F^2 \\ \text{s.t.} \quad & D = B + T + N \end{aligned}$$

$$\Downarrow$$

$$\begin{aligned} \min_{B,T,N} \quad & \mathcal{R}(B) + \lambda \mathcal{S}(T) + \mu \mathcal{G}(N) \\ \text{s.t.} \quad & D = B + T + N \end{aligned}$$

- Unconstrained version

$$\mathcal{L}(B, T, N) = \mathcal{R}(B) + \lambda \mathcal{S}(T) + \mu \mathcal{G}(N) + \frac{\alpha}{2} \|D - B - T - N\|_{\mathrm{F}}^2$$

Algorithm

- Take B-update as an example

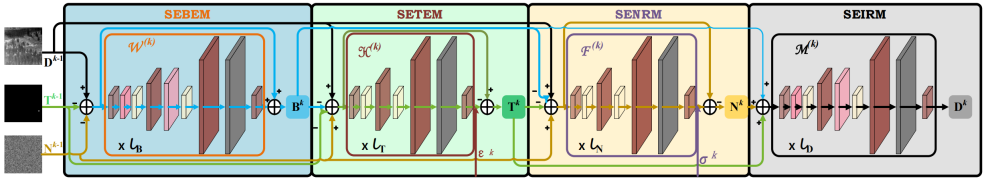
$$B^k = \arg\min_B \ \mathcal{R}(B) + \frac{\alpha}{2}\|D^{k-1} - B - T^{k-1} - N^{k-1}\|_{\mathrm{F}}^2$$

$$\Downarrow$$

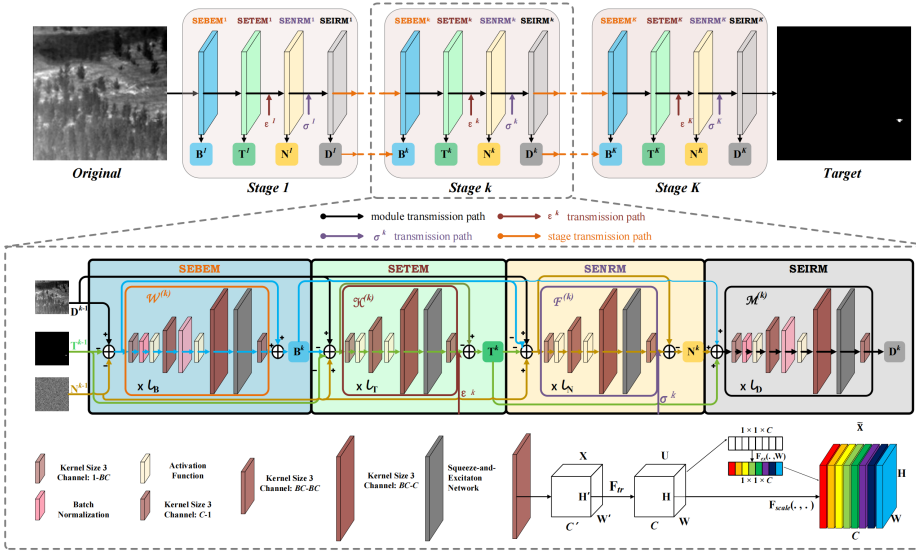
$$B^k = \mathrm{prox}_{\alpha}(D^{k-1} - T^{k-1} - N^{k-1})$$

$$\Downarrow$$

$$B^k = D^{k-1} - T^{k-1} - N^{k-1} + \mathcal{W}^k(D^{k-1} - T^{k-1} - N^{k-1})$$

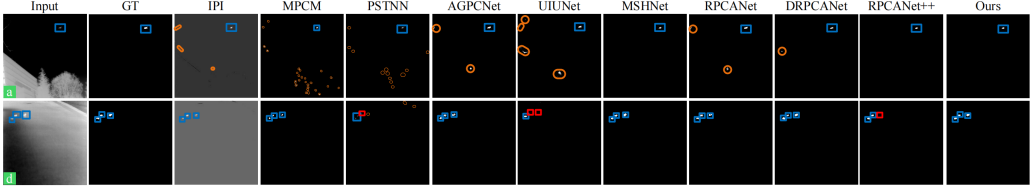


Architecture



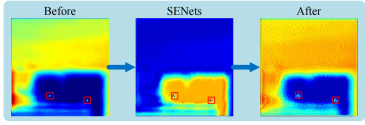
Results

Methods	Params	NUDT-SIRST				SIRST-Aug				IRSTD-1k				Time (s)
		mIoU ↑	F_1 ↑	P_d ↑	F_a ↓	mIoU ↑	F_1 ↑	P_d ↑	F_a ↓	mIoU ↑	F_1 ↑	P_d ↑	F_a ↓	CPU/GPU
IPI	–	34.83	51.49	92.58	7.14	21.90	35.97	80.36	**2.20**	18.67	31.48	78.54	11.11	3.0972/-
MPCM	–	25.96	40.78	78.59	7.91	19.49	33.00	93.58	3.04	14.81	25.93	69.03	6.51	0.0624/-
PSTNN	–	25.46	40.58	78.52	7.95	19.76	33.00	93.40	3.14	14.87	25.89	68.73	6.51	0.2249/-
AGPCNet	12.360M	85.31	92.45	97.90	4.77	72.36	83.83	99.03	35.56	61.00	75.75	89.35	5.34	-/0.0205
UIUNet	50.540M	88.71	94.01	91.43	1.89	71.80	83.59	98.35	28.29	63.06	77.35	**93.60**	6.57	-/0.0317
MSHNet	4.065M	89.99	93.57	96.07	2.63	71.64	84.16	90.78	23.09	64.50	77.55	91.68	4.46	-/0.0245
RPCANet	0.680M	89.31	94.35	97.14	2.87	72.54	84.08	98.21	34.14	63.21	77.45	88.31	4.39	-/0.0096
DRPCANet	1.169M	**93.12**	96.02	98.02	1.95	73.93	85.39	98.12	30.45	63.93	78.15	92.09	4.92	-/0.0101
RPCANet++	4.396M	92.46	96.05	98.05	**1.44**	73.14	84.39	97.36	32.48	64.03	77.26	89.35	**4.28**	-/0.0063
Ours	**0.216M**	92.37	**96.54**	**98.41**	1.79	**74.56**	**85.43**	**99.17**	29.78	**64.68**	**78.55**	89.39	4.66	**-/0.0052**

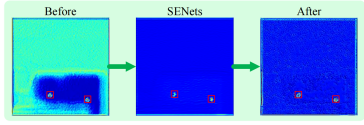


Ablations

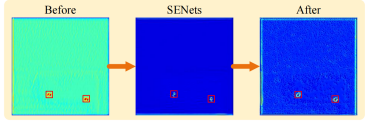
SENets				NUDT-SIRST				SIRST-Aug				IRSTD-1k			
SEBEM	SETEM	SENRM	SEIRM	mIoU ↑	F_1 ↑	P_d ↑	F_a ↓	mIoU ↑	F_1 ↑	P_d ↑	F_a ↓	mIoU ↑	F_1 ↑	P_d ↑	F_a ↓
✗	✗	✗	✗	73.56	78.45	79.36	8.56	60.75	70.28	81.24	43.67	50.26	61.34	70.57	15.95
✓	✗	✗	✗	80.57	81.39	86.35	5.68	65.96	75.37	86.99	39.82	55.84	65.26	76.36	11.12
✓	✓	✗	✗	88.36	89.45	90.18	4.00	70.17	80.78	93.17	34.78	60.56	72.58	83.70	8.10
✓	✓	✓	✗	91.14	96.06	97.18	2.05	73.27	84.45	98.07	27.73	63.58	77.53	88.71	3.95
✓	✓	✓	✓	**92.37**	**96.54**	**98.41**	**1.79**	**74.56**	**85.43**	**99.17**	**29.78**	**64.68**	**78.55**	**89.39**	**4.66**



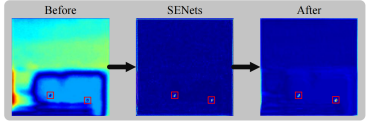
(a) SEBEM



(b) SETEM



(c) SENRM



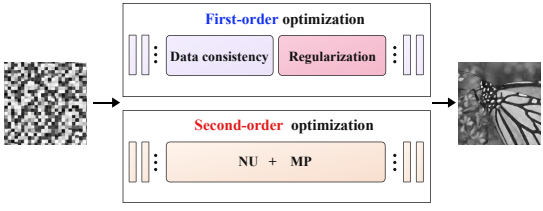
(d) SEIRM

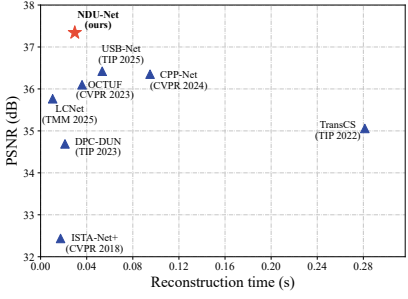
Part III

- He-Xiu, Newton Deep Unfolding for Compressed Sensing, submitted to AAAI

$$\min_{x\in\mathbb{R}^n} \underbrace{\frac{1}{2}\|Ax-b\|^2}_{f(x)} + \lambda\mathcal{R}(x)$$

- Iterative optimization: ISTA, FISTA, AMP, PDHG, ADMM, SSNAL, NHTP
- Deep unrolling: LISTA, ALISTA, ISTA-Net, ISTA-Net+, LDAMP, ADMM-CSNet





Step 1

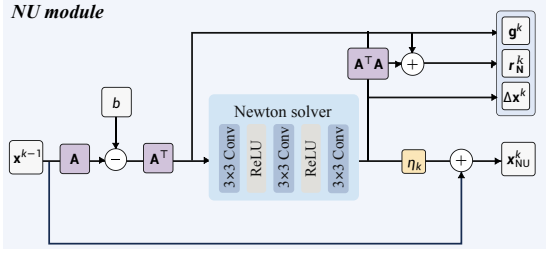
- *(ISTA)* Compute gradient

$$r^{k+1} = x^k - \rho \nabla f(x^k)$$

- *(NDU-Net)* The Newton update (NU) satisfies

$$\begin{cases} \nabla^2 f(x^k)\Delta x^k + \nabla f(x^k) = 0 \\ x^k_{\mathrm{NU}} \leftarrow x^k + \eta_k \Delta x^k \end{cases}$$

Define the Newton residual as $r_N^k = A^\top A \Delta x^k + \nabla f(x^k)$



Step 2

- *(ISTA)* Compute soft-thresholding

$$x^{k+1} = \texttt{Prox}_{\lambda\|\cdot\|_1}(x^k - \rho\nabla f(x^k))$$

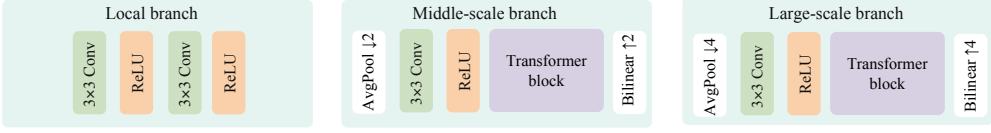
- *(NDU-Net)* Extract multi-scale prior (MP) information

$$y_{MS}^k = \mathcal{P}_{\mathrm{MS}}^k(x_{\mathrm{NU}}^k)$$

Encode Newton guidance information

$$y_N^k = \mathcal{E}_N^k(g^k, r_N^k, \Delta x^k)$$

$$\Rightarrow \quad y_{\mathrm{guided}}^k = y_{MS}^k + y_N^k$$



Step 2

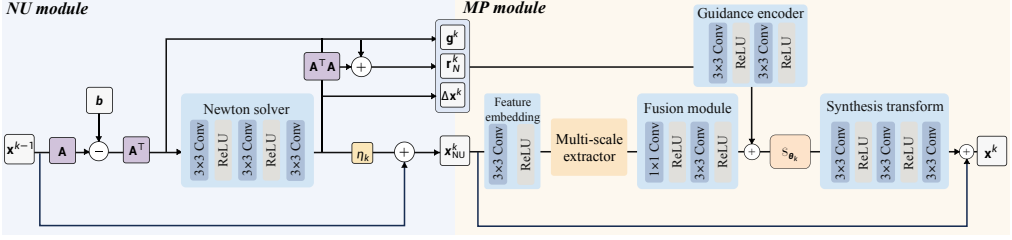
- *(NDU-Net)* Apply a learnable soft-thresholding operation

$$y^k_{\text{shrink}} = \mathcal{S}_{\theta_k}(y^k_{\text{guided}})$$

$$\mathcal{S}_{\theta_k}(z) = \text{sign}(z) \odot \max(|z| - \theta_k, 0)$$

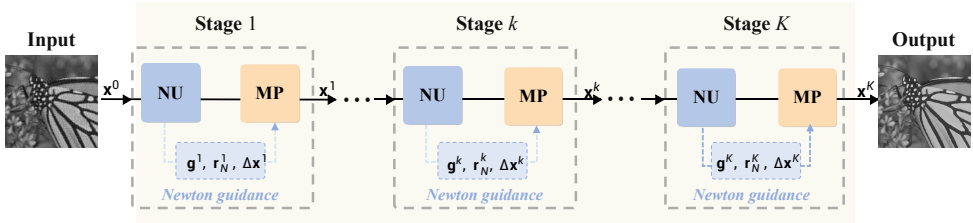
Transform back to the image domain

$$r^k_{\text{prior}} = \mathcal{H}^k(y^k_{\text{shrink}})$$



Architecture

- Overall architecture



- In the k-th stage

$$x^{k+1} = x^k_{\text{NU}} \text{ (Step 1)} + r^k_{\text{prior}} \text{ (Step 2)}$$

Training loss

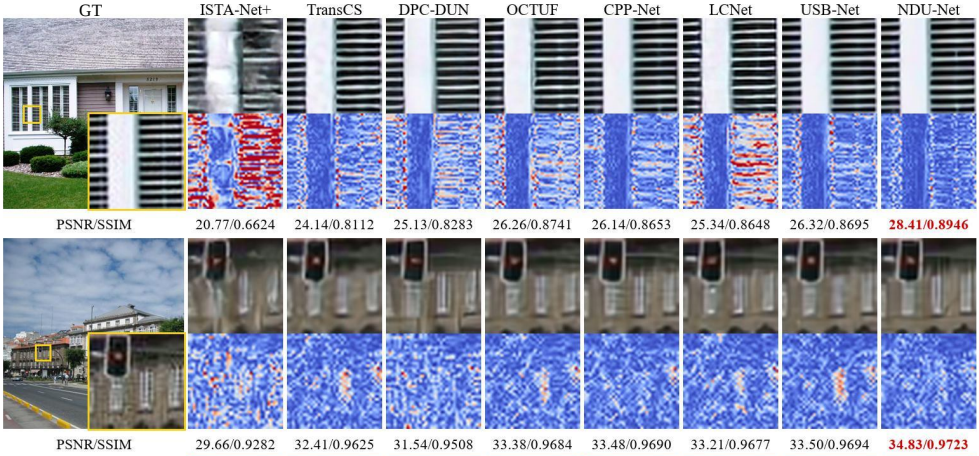
$$\ell_{\text{total}} = \ell_{\text{rec}} + \sigma \ell_{\text{Newton}}$$

$$= \frac{1}{N}\|x^K - x\|_1 + \sigma \left(\frac{1}{K} \sum_{k=0}^{K-1} \left(\frac{1}{N} \|r_N^k\|_1 \right) \right)$$

Results

Datsets	Methods	CS ratios 0.10	0.25	0.40	0.50	Average
LIVE29	ISTA-Net+ (CVPR'18)	25.05/0.7104	29.42/0.8648	32.55/0.9275	34.55/0.9514	30.39/0.8635
	TransCS (TIP'22)	27.66/0.8196	32.05/0.9251	35.51/0.9633	37.74/0.9767	33.24/0.9212
	DPC-DUN (TIP'23)	26.83/0.7812	31.17/0.9008	34.43/0.9477	36.43/0.9649	32.22/0.8987
	OCTUF (CVPR'23)	28.23/0.8345	32.69/0.9327	36.19/0.9671	38.52/0.9795	33.91/0.9285
	CPP-Net (CVPR'24)	28.46/0.8414	32.82/0.9343	36.20/0.9672	38.37/0.9790	33.96/0.9305
	LCNet (TMM'25)	28.19/0.8349	32.64/0.9325	36.08/0.9667	38.34/0.9791	33.81/0.9283
	USB-Net (TIP'25)	28.48/0.8420	32.87/0.9348	36.29/0.9676	38.46/0.9793	34.03/0.9309
	NDU-Net	**29.62/0.8517**	**34.07/0.9399**	**37.55/0.9708**	**39.62/0.9812**	**35.22/0.9359**
OST300	ISTA-Net+ (CVPR'18)	24.78/0.6896	28.53/0.8433	31.34/0.9116	33.20/0.9396	29.46/0.8460
	TransCS (TIP'22)	27.31/0.8019	31.07/0.9096	34.17/0.9534	36.25/0.9701	32.20/0.9088
	DPC-DUN (TIP'23)	26.25/0.7561	29.93/0.8792	32.81/0.9335	34.69/0.9554	30.92/0.8811
	OCTUF (CVPR'23)	27.77/0.8148	31.60/0.9175	34.61/0.9572	36.69/0.9726	32.67/0.9155
	CPP-Net (CVPR'24)	27.93/0.8207	31.67/0.9185	34.66/0.9573	36.68/0.9724	32.74/0.9172
	LCNet (TMM'25)	27.73/0.8151	31.53/0.9169	34.53/0.9568	36.60/0.9723	32.60/0.9153
	USB-Net (TIP'25)	27.95/0.8214	31.71/0.9191	34.71/0.9578	36.71/0.9726	32.77/0.9177
	NDU-Net	**29.22/0.8314**	**33.00/0.9239**	**36.01/0.9604**	**37.94/0.9739**	**34.04/0.9224**

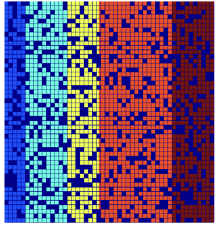
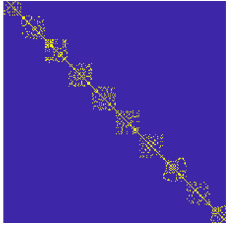
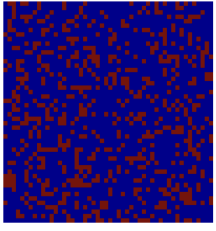
Visualization



Part IV

- Liu-Lin-Yan et al, Robust Recovery of Subspace Structures by Low-Rank Representation, IEEE TPAMI, 2013

$$\min_{Z,E} \ \operatorname{rank}(Z) + \lambda\|E\|_0 \quad \text{s.t.} \ \ X = XZ + E \qquad \Rightarrow \qquad \min_{Z,E} \ \|Z\|_* + \lambda\|E\|_1 \quad \text{s.t.} \ \ X = XZ + E$$

- Board applications: clustering, classification, denoising, detection, fusion
- Related to low-rank regularization, low-rank approximation, low-rank decomposition

Model

- Chen-Wang-Xiu et al, Rethinking Low-Rank Representation with Deep Semi-Smooth Newton, PRCV, 2026

$$\begin{aligned} \min_{Z,E} \ & \|Z\|_* + \lambda\|E\|_1 \\ \text{s.t.} \ & X = XZ + E \end{aligned} \quad \Rightarrow \quad \begin{aligned} \min_{Z,E} \ & \mathcal{R}(Z) + \lambda\mathcal{S}(E) \\ \text{s.t.} \ & X = XZ + E \end{aligned}$$

- Augmented Lagrangian method (ALM)

$$\begin{aligned} \min_{J,E,Z} \ & \mathcal{R}(J) + \lambda\mathcal{S}(E) \\ \text{s.t.} \ & X = XZ + E, \ Z = J \end{aligned}$$

which derives

$$\begin{aligned} \mathcal{L}_\beta(J, E, Z, Y_1, Y_2) &= \mathcal{R}(J) + \lambda\mathcal{S}(E) + \langle Y_1, Z - J\rangle \\ &+ \frac{\beta}{2}\|Z - J\|_F^2 + \langle Y_2, X - XZ - E\rangle + \frac{\beta}{2}\|X - XZ - E\|_F^2 \end{aligned}$$

Algorithm

- Update variables according to

$$\begin{cases} (J, E, Z) \leftarrow \arg\min_{J,E,Z} \ \mathcal{L}_\beta(J, E, Z, Y_1, Y_2) \\ Y_1 \leftarrow Y_1 + \beta(Z - J) \\ Y_2 \leftarrow Y_2 + \beta(X - XZ - E) \end{cases}$$

- First-order optimality conditions

$$\begin{cases} 0 \in \partial\mathcal{R}(J) - Y_1 + \beta(J - Z) \\ 0 \in \lambda\,\partial\mathcal{S}(E) - Y_2 - \beta(X - XZ - E) \\ 0 = Y_1 + \beta(Z - J) - X^\top Y_2 - \beta X^\top (X - XZ - E) \end{cases}$$

which derives

$$J = \mathtt{Prox}_{1/\beta,\mathcal{R}}\left(Z + Y_1/\beta\right), \quad E = \mathtt{Prox}_{\lambda/\beta,\mathcal{S}}\left(X - XZ + Y_2/\beta\right)$$

SSN

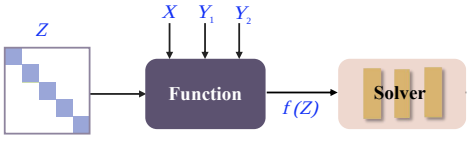
- Denote the nonlinear equation

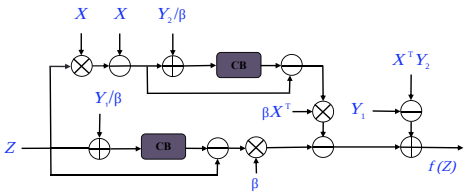
$$
\begin{aligned}
f(Z) = Y_1 + \beta\left(Z - \mathtt{Prox}_{1/\beta,\mathcal{R}}(Z + Y_1/\beta)\right) - X^\top Y_2 \\
- \beta X^\top \left(X - XZ - \mathtt{Prox}_{\lambda/\beta,\mathcal{S}}(X - XZ + Y_2/\beta)\right)
\end{aligned}
$$

$$
\Downarrow
$$

$$
\begin{cases}
g(Z)\Delta Z = -f(Z) \\
Z \leftarrow Z + \eta \Delta Z
\end{cases}
$$

- Detailed structure: NS (Newton solver), CB (convolutional block)



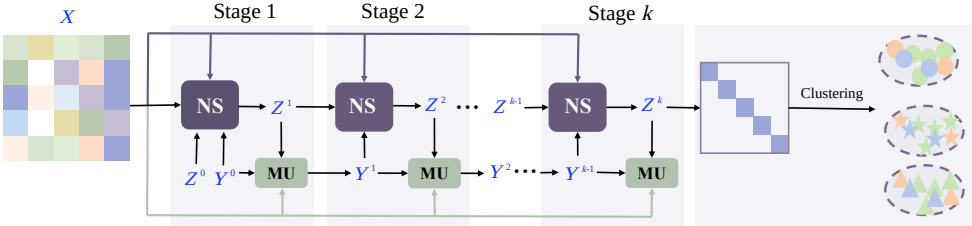


Architecture

- Training loss

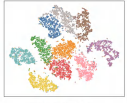
$$\ell_{\text{total}} = \ell_{\text{self}} + \sigma_1 \ell_{\text{fun}} + \sigma_2 \ell_{\text{reg}}$$

$$= \left(\|X - XZ^K - E^K\|_F^2 + \gamma \|E^K\|_1 \right) + \sigma_1 \left(\frac{1}{N} \sum_{l=1}^{N} \|f(Z^K)\|_1 \right) + \sigma_2 \text{tr}(L_{Z^K})$$



Results

Methods	Datasets								Average	
	AR	FRDUE	MNIST	USPS	GLOMA	ISOLET	MSTAR	TDT2	ACC	Time
IDR (TKDE'23)	72.64%	95.31%	58.61%	79.80%	66.00%	67.65%	84.50%	84.10%	76.08%	225.80
TS1-LLRR (KBS'24)	76.46%	88.64%	49.65%	79.48%	61.60%	65.64%	87.61%	84.51%	74.20%	265.10
JSSC (ESWA'25)	78.09%	91.31%	46.78%	69.22%	50.00%	66.61%	65.99%	59.37%	65.92%	358.60
ISC (TPAMI'26)	54.42%	88.19%	44.02%	65.30%	60.00%	81.02%	43.13%	51.75%	60.98%	118.50
DSC-Net (NeurIPS'17)	45.19%	82.53%	66.07%	90.00%	70.00%	41.41%	82.53%	12.45%	61.27%	23.23
PRO-DSC (ICLR'25)	63.75%	**96.09%**	45.39%	67.60%	59.40%	52.00%	66.41%	88.08%	67.34%	**0.03**
DCSSC (Neuro'25)	83.68%	80.46%	87.25%	90.30%	84.00%	86.00%	91.85%	78.35%	85.24%	0.85
LRRSSN-Net	**89.95%**	96.02%	**93.43%**	**99.40%**	**98.00%**	**97.72%**	**99.72%**	**96.40%**	**96.33%**	0.21



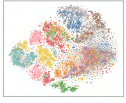
(a) IDR



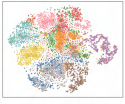
(b) TS1-LLRR



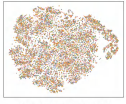
(c) JSSC



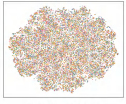
(d) ISC



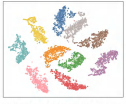
(e) DSC-Net



(f) PRO-DSC



(g) DCSSC



(h) LRRSSN-Net

Remarks

- Chen-Wang-Xiu et al, LLM-IMVC: Large Language Model-Guided Deep Unrolling for Incomplete Multi-View Clustering, working in process

$$\min_{Z,E} \ \mathcal{R}(Z) + \lambda \mathcal{S}(E) \quad \text{s.t.} \ \ X = XZ + E \qquad \Rightarrow \qquad \min_{Z_v, E_v} \ \sum_{v=1}^{V} \mathcal{R}(Z_v) + \lambda \mathcal{S}(E_v) \quad \text{s.t.} \ \ \mathcal{P}_\Omega(X_v) = \mathcal{P}_\Omega(X_v Z_v + E_v)$$

Methods	HandWrite	BDGP	MSRC	COIL20	Caltech5V	USPS	Average
LMVSC (AAAI'20)	67.1%	55.6%	35.2%	69.2%	46.1%	56.0%	54.9%
SFMC (TPAMI'22)	75.7%	37.8%	60.0%	74.8%	46.8%	98.9%	65.7%
FPMVS-CAG (TIP'22)	82.3%	55.6%	80.5%	60.2%	80.9%	78.3%	73.0%
FDAGF (AAAI'23)	82.3%	52.5%	73.9%	74.0%	81.3%	56.1%	70.0%
RCAGL (TKDE'24)	79.4%	51.8%	77.6%	65.1%	83.6%	60.9%	69.7%
ALPC (AAAI'25)	63.2%	92.2%	64.3%	76.4%	46.1%	50.2%	65.4%
TLRLF4 (TPAMI'25)	76.4%	92.7%	—	76.3%	75.5%	99.2%	84.0%
LLM-IMVC	**99.9%**	**100.0%**	**88.1%**	**77.2%**	**100.0%**	**99.9%**	**94.2%**

Part V

- LLM for Optimization & Optimization for LLM

对于提出的联邦 PCA 模型，请设计 ADMM 算法，提供代码，并分析收敛性和复杂度
\begin {equation}\label {eq:proposed_method}
\begin {array}{cl}
\displaystyle \min_{{\{\mathbf {W}_i\}, {\mathbf {S} i}, \mathbf {V}} \quad & \sum\limits {i=1}^d \big \{ (\mathbf {I} - \mathbf {W} i \mathbf {W}{i}^{T})(\mathbf {X}_i - \mathbf {S}_i) \|_F^2
& + \alpha \|\mathbf{S}_i\|_1 + \beta \|\mathbf{W}i\|{2,1}\big\} [6pt]
\textrm{s.t.} & \mathbf{W}_i = \mathbf{V}, \quad \mathbf{W}i^{T} \mathbf{W}{i}=\mathbf{I}, \quad \forall i \in [d],
\end{array}
\end{equation}

联邦鲁棒 PCA：ADMM 完整设计 + Python 可运行代码 + 收敛性分析 + 计算复杂度

整体框架概述

符号约定

- d：客户端数量；第 i 方数据 $\boldsymbol{X}_i \in \mathbb{R}^{n_i \times p}$，$n_i$：每方样本数，$p$：特征维度
- $\boldsymbol{W}_i \in \mathbb{R}^{p \times k}$：客户端局部投影矩阵，满足 Stiefel 正交约束 $\boldsymbol{W}_i^\top \boldsymbol{W}_i = \boldsymbol{I}_k$
- $\boldsymbol{V} \in \mathbb{R}^{p \times k}$：全局共享公共投影矩阵（联邦耦合变量）
- $\boldsymbol{S}_i \in \mathbb{R}^{n_i \times p}$：稀疏噪声 / 异常矩阵，$\ell_1$ 正则去除相差
- $\|\boldsymbol{W}_i\|_{2,1}$：行稀疏正则，让投影基具备行稀疏性
- $\alpha > 0$：稀疏权重；$\beta > 0$：行稀疏权重；$\rho > 0$：ADMM 增广拉格朗日惩罚系数

约束：$\forall i, \boldsymbol{W}_i = \boldsymbol{V},\ \boldsymbol{W}_i^\top \boldsymbol{W}_i = \boldsymbol{I}$

原目标：

$$\min_{\{\boldsymbol{W}_i, \boldsymbol{S}_i\}, \boldsymbol{V}} \sum_{i=1}^{d} \left\{ \left\| (\boldsymbol{I} - \boldsymbol{W}_i \boldsymbol{W}_i^\top)(\boldsymbol{X}_i - \boldsymbol{S}_i) \right\|_F^2 + \alpha \|\boldsymbol{S}_i\|_1 + \beta \|\boldsymbol{W}_i\|_{2,1} \right\}, \quad \text{s.t. } \boldsymbol{W}_i = \boldsymbol{V}, \boldsymbol{W}_i^\top \boldsymbol{W}_i = \boldsymbol{I}$$

一、ADMM 算法推导

1. 增广拉格朗日函数

引入对偶乘子 $\boldsymbol{\Lambda}_i \in \mathbb{R}^{p \times k}$，增广拉格朗日：

$$\mathcal{L}_\rho(\{\boldsymbol{W}_i, \boldsymbol{S}_i\}, \boldsymbol{V}, \{\boldsymbol{\Lambda}_i\}) = \sum_{i=1}^{d} \left[\left\| (\boldsymbol{I} - \boldsymbol{W}_i \boldsymbol{W}_i^\top)(\boldsymbol{X}_i - \boldsymbol{S}_i) \right\|_F^2 + \alpha \|\boldsymbol{S}_i\|_1 + \beta \|\boldsymbol{W}_i\|_{2,1} + \langle \boldsymbol{\Lambda}_i, \boldsymbol{W}_i - \boldsymbol{V} \rangle + \frac{\rho}{2} \|\boldsymbol{W}_i - \boldsymbol{V}\|_F^2 \right]$$

ADMM 四块交替迭代：

1. 客户端并行更新 $\boldsymbol{S}_i$
2. 客户端并行求解带 Stiefel 约束的 $\boldsymbol{W}_i$
3. 服务器聚合更新全局 $\boldsymbol{V}$
4. 更新对偶变量 $\boldsymbol{\Lambda}_i$

2. 子问题闭式 / 数值解

(1) $\boldsymbol{S}_i$ 更新（逐元素软阈值）

固定其余变量，令 $\boldsymbol{Y}_i = \boldsymbol{X}_i - \boldsymbol{S}_i$，$\boldsymbol{P}_i = \boldsymbol{I} - \boldsymbol{W}_i \boldsymbol{W}_i^\top$ 幂等投影阵：

$$\min_{\boldsymbol{S}_i} \|\boldsymbol{P}_i(\boldsymbol{X}_i - \boldsymbol{S}_i)\|_F^2 + \alpha \|\boldsymbol{S}_i\|_1$$

闭式解：

$$\boldsymbol{S}_i = \mathcal{S}_\alpha\left(\boldsymbol{X}_i - \boldsymbol{W}_i \boldsymbol{W}_i^\top \boldsymbol{X}_i\right)$$

软阈值算子：$\mathcal{S}_\tau(z) = \operatorname{sign}(z) \max(|z| - \tau, 0)$

python

```python
import numpy as np

# ==================== 算子工具函数 ====================
def soft_threshold(x, tau):
    """L1软阈值"""
    return np.sign(x) * np.maximum(np.abs(x) - tau, 0.0)

def row_21_prox(W, beta):
    """L2,1行近端算子：每行L2范数收缩"""
    p, k = W.shape
    for r in range(p):
        norm_r = np.linalg.norm(W[r, :], ord=2)
        if norm_r > 1e-8:
            scale = max(1 - beta / norm_r, 0.0)
            W[r, :] *= scale
    return W

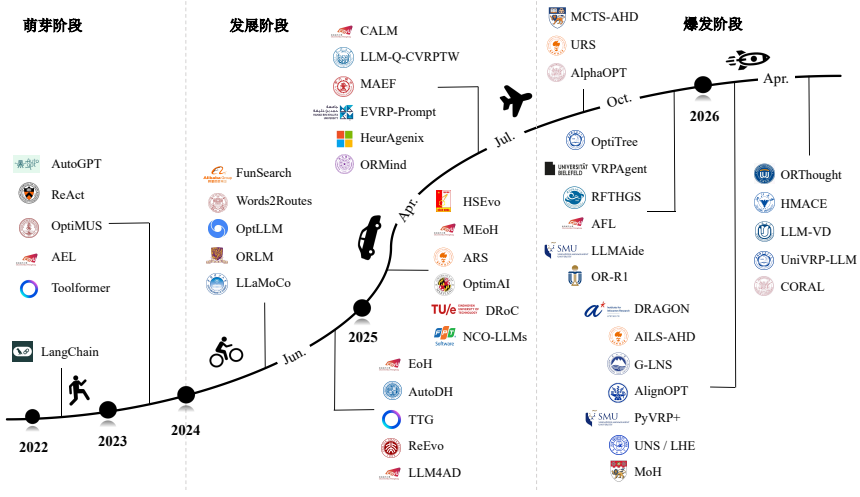
def stiefel_retraction(W):
    """SVD投影至Stiefel流形 W^T W = I"""
    U, S, Vt = np.linalg.svd(W, full_matrices=False)
    return U @ Vt

def stiefel_pgd(C, V_ref, Lambda, rho, beta, lr=1e-2, inner_it
    """
    Stiefel流形近端梯度下降求解W
    min -tr(W^T C) + 0.5*rho||W - (V_ref - Lambda/rho)||_F^2 +
    """
    p, k = C.shape[0], V_ref.shape[1]
    W = np.random.randn(p, k)
    W = stiefel_retraction(W)
    target = V_ref - Lambda / rho

    for _ in range(inner_iter):
        # 欧氏梯度
        grad = -C + rho * (W - target)
        W = W - lr * grad
        # L2,1近端
        W = row_21_prox(W, beta * lr)
        # 流形回缩正交化
        W = stiefel_retraction(W)
    return W
```

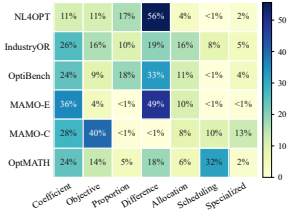
VRP

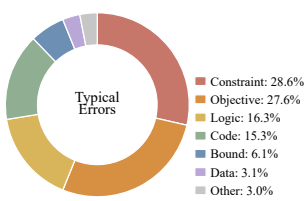
- Xiu-Shen-Zhu et al, A Survey of Large Language Models for Vehicle Routing Problems (in Chinese), CTA, minor revision

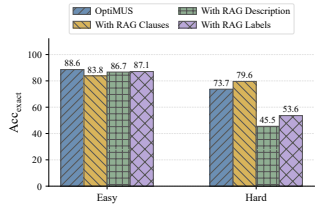


GraphRAG

- Edge-Trinh-Cheng et al, From Local to Global: A GraphRAG Approach to Query-Focused Summarization arXiv:2404.16130
- Gutiérrez-Shu-Gu et al, HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models, NeurIPS, 2024
- Guo-Xia-Yu et al, LightRAG: Simple and Fast Retrieval-Augmented Generation, EMNLP, 2025
- Zhu-Huang-Wang et al, Graph-Based Approaches and Functionalities in Retrieval-Augmented Generation: A Comprehensive Survey, ACM Computing Surveys, 2026







Example

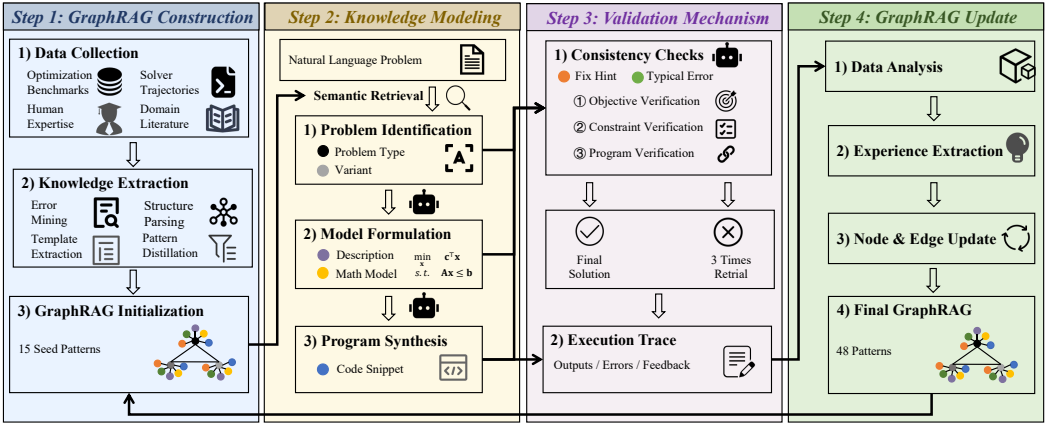
- Problem Type: Vehicle routing problem (VRP)
- Variant: Capacitated vehicle routing problem (CVRP)
- Description: Design vehicle delivery routes to serve all customers with minimal travel cost under vehicle capacity constraints
- Math Model:

$$
\begin{aligned}
\min_{x} \quad & \sum_{k \in K} \sum_{i \in V} \sum_{j \in V} c_{ij} x_{ijk} \\
\text{s.t.} \quad & \sum_{k \in K} \sum_{j \in V} x_{ijk} = 1, && \forall i \in C \\
& \sum_{i \in C} d_i \sum_{j \in V} x_{ijk} \leq Q, && \forall k \in K \\
& x_{ijk} \in \{0, 1\}, && \forall i, j \in V,\ k \in K
\end{aligned}
$$

- Code Snippet:
```
x = addVars(V, V, K, vtype=GRB.BINARY)
min sum(c[i,j] * x[i,j,k] for i in V for j in V for k in K)
addConstrs(each customer visited once), addConstrs(vehicle capacity limits)
```
- Typical Error: Missing customer coverage, ignoring vehicle capacity, disconnected subtours
- Fix Hint: Visit each customer once, add capacity constraints, eliminate subtours

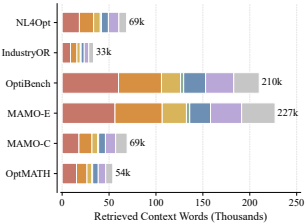
Workflow

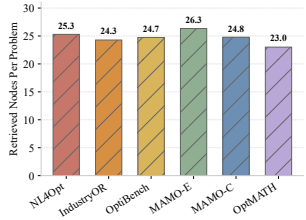
- Xiu-Li-Huang et al, OptGraph: Large Language Models Enhanced Evolutionary Optimization Via Graph Retrieval-Augmented Generation, submitted to ICLR

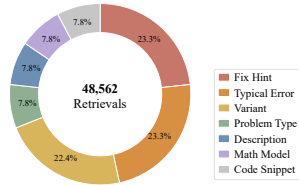


Accuracy

Methods	NL4OPT	IndustryOR	OptiBench	MAMO-E	MAMO-C	OptMATH
GPT-5.1	72.9%/78.0%	43.0%/51.0%	58.0%/65.8%	81.6%/85.8%	47.8%/58.1%	27.7%/30.7%
Gemini-3.1-Pro	73.4%/78.5%	42.0%/50.0%	58.2%/66.0%	82.1%/86.1%	46.8%/57.1%	27.1%/30.1%
DeepSeek-V3.2	72.4%/77.6%	44.0%/50.0%	57.4%/65.1%	81.0%/85.0%	48.3%/57.6%	26.5%/30.1%
Qwen3.6-Plus	69.6%/74.8%	42.0%/49.0%	55.4%/62.6%	78.0%/81.8%	46.3%/55.7%	27.1%/28.9%
CoE	77.1%/83.6%	51.0%/61.0%	65.5%/73.2%	93.5%/93.8%	67.0%/70.3%	40.4%/43.4%
OptiMUS	86.9%/93.9%	54.0%/60.0%	63.0%/72.1%	87.8%/89.0%	52.2%/70.4%	38.5%/41.3%
OptiTree	84.1%/92.5%	64.0%/71.0%	62.5%/71.9%	90.7%/91.2%	65.4%/72.8%	31.3%/33.7%
Lean-LLM-OPT	86.5%/94.9%	55.0%/63.0%	61.8%/70.3%	93.8%/94.2%	57.6%/70.0%	44.5%/45.2%
OptGraph (Ours)	**86.9%/95.3%**	**71.0%/81.0%**	**66.0%/78.2%**	**93.9%/94.4%**	**76.4%/89.2%**	**58.4%/60.8%**

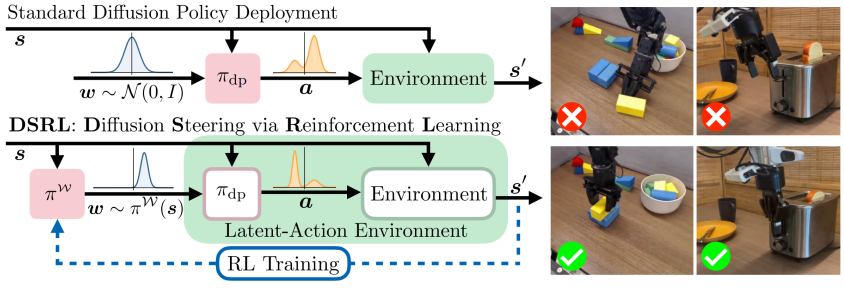






Part VI

- Wagenmaker-Nakamoto-Zhang et al, Steering Your Diffusion Policy with Latent Space Reinforcement Learning, CoRL, 2025



- How to improve the action representation?
- How to develop a lightweight residual modulation mechanism?

Methodology

- Zhang-Sun-Xiu, Beyond Noise Steering: Dual-Latent Space Reinforcement Learning for Generative Robot Policy, submitted to ICRA

$$(z_t, u_t) \sim \pi_\theta(\cdot \mid o_t), \;\; f_t = \mathcal{A}_\omega(u_t)$$

$$\Downarrow$$

$$a_t = G_\phi(o_t, z_t; f_t)$$

- How to develop a lightweight residual modulation mechanism?

$$H_{t,\kappa}^{(l)} = \mathcal{T}_\phi^{(l)}(\widetilde{H}_{t,\kappa}^{(l-1)}; e_t, \kappa), \;\; \delta_t = \lambda_{\text{inj}}\, \mathcal{B}(f_t)$$

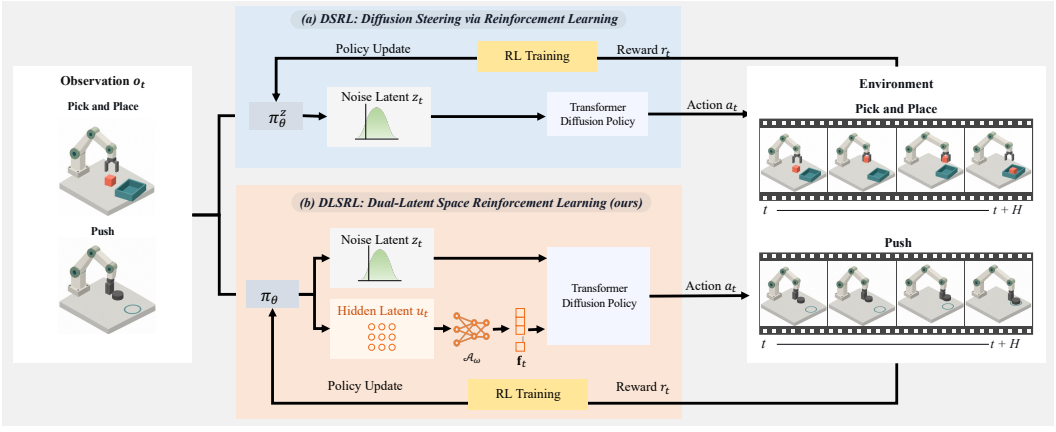
$$\Downarrow$$

$$\widetilde{H}_{t,\kappa}^{(l)} = H_{t,\kappa}^{(l)} + \delta_t$$

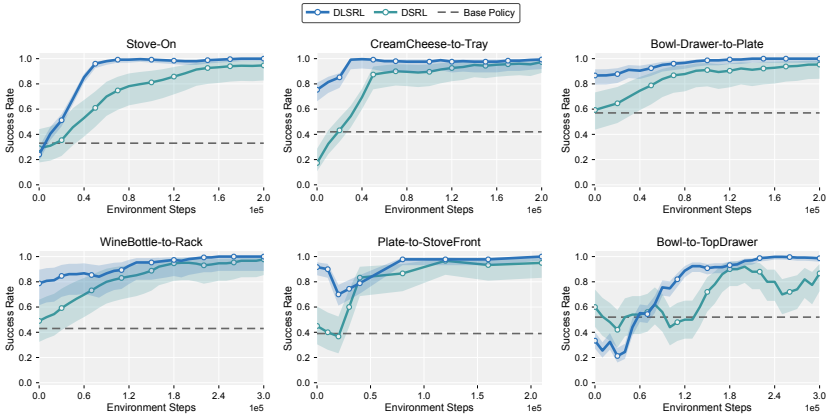
- Latent-space optimization

$$\mathcal{L}_{\text{actor}}(\theta, \omega) = E_{o\sim\mathcal{D},(z,u)\sim\pi_\theta(\cdot|o)}\Big[\alpha \log \pi_\theta^z(z \mid o) - Q_\nu^{\text{lat}}(o, \xi)\Big] + \mathcal{R}_{\text{aux}}$$

Architecture



VLA



Methods	Stove-On	CreamCheese-to-Tray	Bowl-Drawer-to-Plate	WineBottle-to-Rack	Plate-to-StoveFront	Bowl-to-TopDrawer
DSRL	88.34 ± 16.42	203.44 ± 12.58	125.80 ± 29.18	113.26 ± 8.97	101.24 ± 15.82	226.90 ± 46.04
DLSRL (Ours)	$\mathbf{65.90 \pm 1.12}$	$\mathbf{157.20 \pm 5.77}$	$\mathbf{97.06 \pm 0.22}$	$\mathbf{95.96 \pm 0.65}$	$\mathbf{80.29 \pm 3.42}$	$\mathbf{180.16 \pm 11.82}$
Reduction	25.40%	22.73%	22.85%	15.27%	20.69%	20.60%

Outline

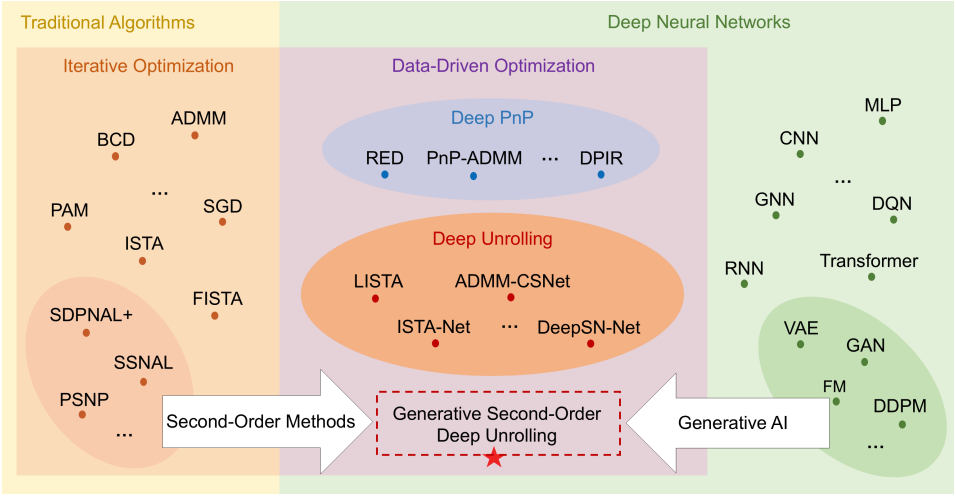
Introduction

Recent Advances

Future Work

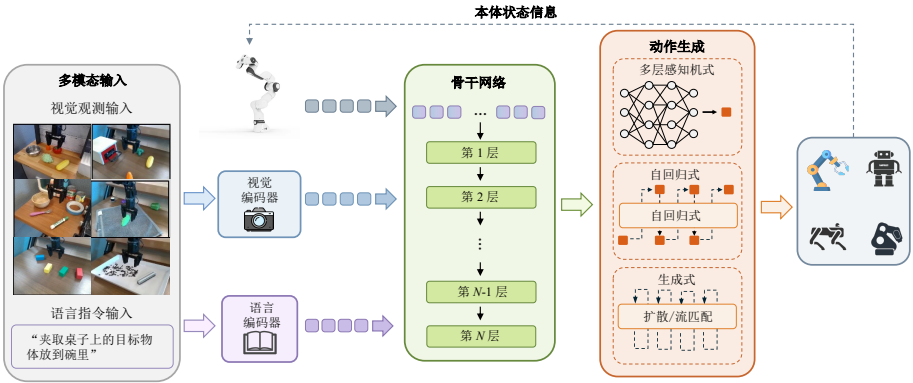
Future Work

- Second-order Optimization + Deep Unrolling + Generative AI



Future Work

- Xiu-Xu-Zhu et al, Review of Compression Methods for Vision-Language-Action Models (in Chinese), submitted to Acta Automatica Sinica



Thank you for your attention!

xcxiu@shu.edu.cn